

1. Create an S3 bucket and upload some objects to S3.

1. Create S3 Bucket & Upload Objects

- Created an Amazon S3 bucket.
- Configured bucket ownership and security settings.
- Enabled Block Public Access.
- Enabled versioning.
- Enabled server-side encryption.
- Uploaded multiple objects to the bucket.
- Verified uploaded objects in S3.

Amazon S3

Upload: status

After you navigate away from this page, the following information is no longer available.

Summary

Destination s3://syed-s3-lab-2026-001	Succeeded 2 files, 29.0 B (100.00%)	Failed 0 files, 0 B (0%)
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Files and folders | Configuration

Files and folders (2 total, 29.0 B)

Find by name

Name	Folder	Type	Size	Status	Error
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2. Deploy a static website in the S3 bucket.

Created an S3 bucket for static website hosting.

Uploaded `index.html`.

Enabled static website hosting.

Configured `index.html` as the index document.

Tested the static website.

← ↻ ⚠ Not secure syed-s3-lab-2026-001.s3-website-us-east-1.amazonaws.com

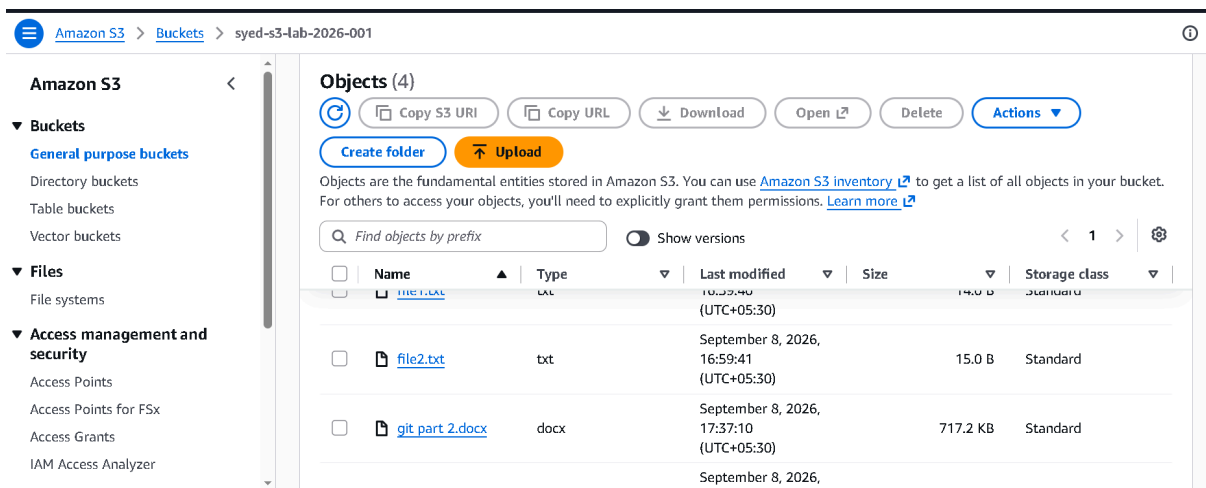
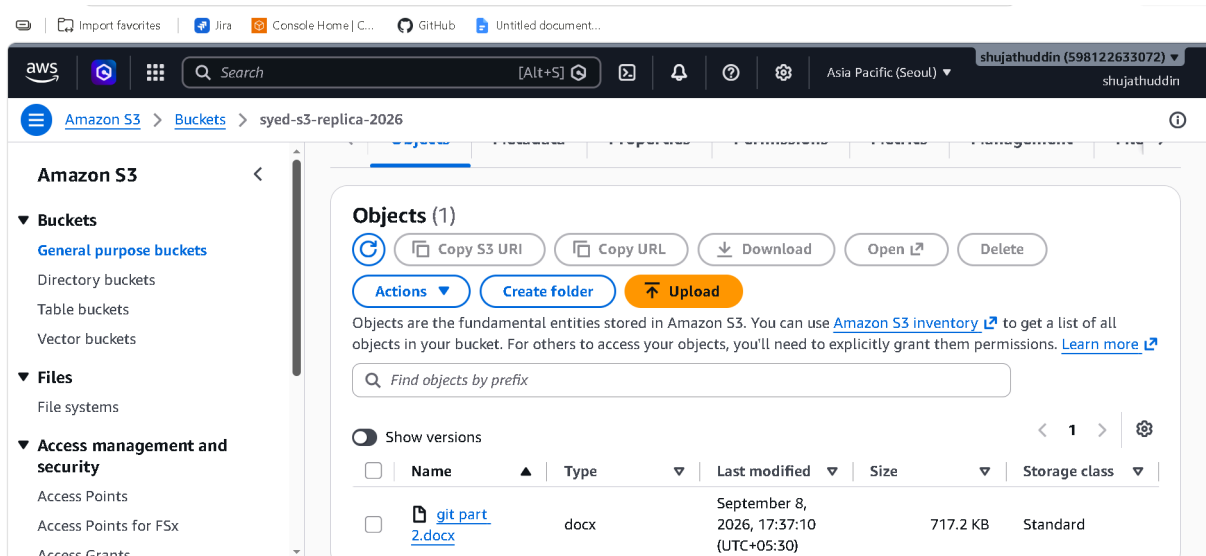
🔖 Import favorites | 🏠 Jira | 📁 Console Home | C... | 🌐 GitHub | 📄 Untitled document...

Welcome to My S3 Static Website

This website is hosted using Amazon S3.

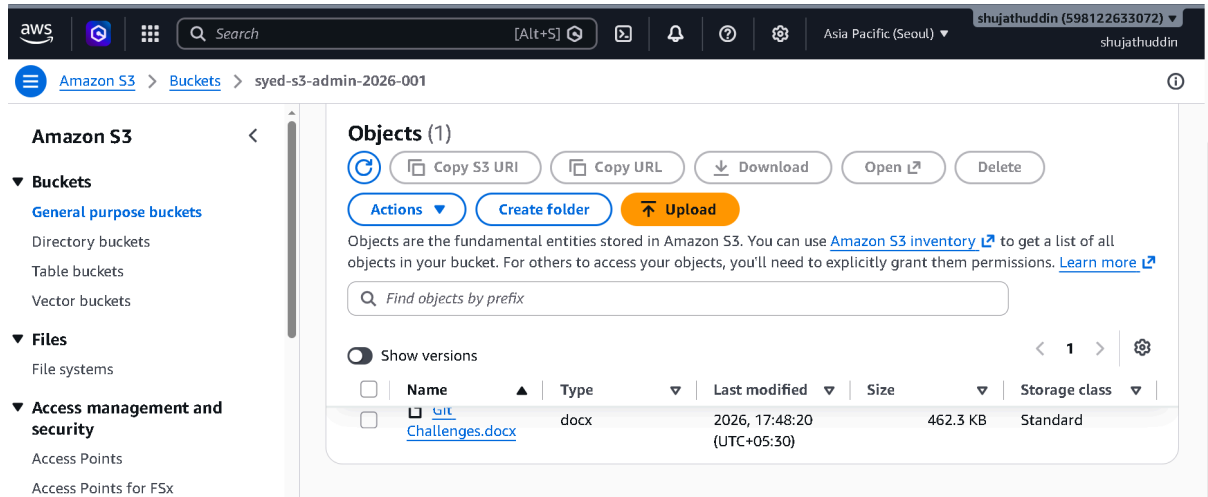
3. Enable cross-region replication on S3 buckets.

- Enabled versioning on the S3 buckets.
- Created a replication rule.
- Configured source bucket in **Seoul (ap-northeast-2)**.
- Configured destination bucket in **N. Virginia (us-east-1)**.
- Created the required IAM replication role.
- Uploaded a new object to test replication.
- Verified replication.



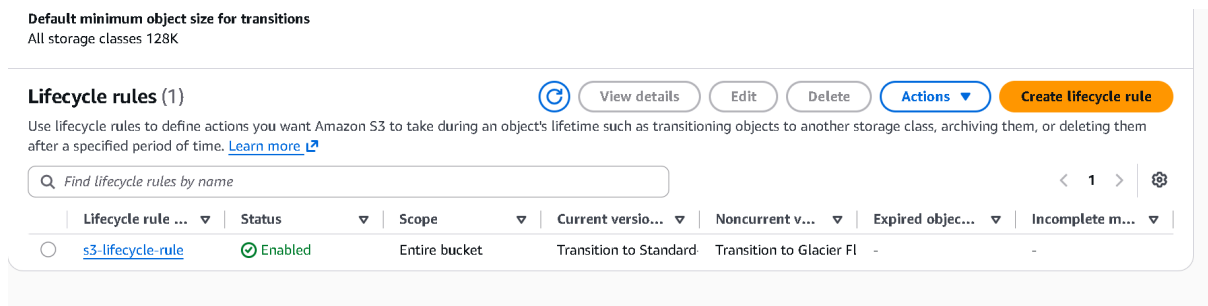
4. Configure a bucket policy so only the Admin user can see the objects of the S3 bucket.

- Created an IAM user named **Admin**.
- Configured S3 bucket policy.
- Allowed **s3:ListBucket** permission.
- Allowed **s3:GetObject** permission.
- Restricted access to the Admin user.
- Kept Block Public Access enabled.
- Successfully saved the bucket policy.



5. Set up lifecycle policies to automatically transition or delete objects based on specific criteria.

1. Created an S3 lifecycle rule.
2. Applied the rule to bucket objects.
3. Configured transition to **Standard-IA** after **30 days**.
4. Configured transition to **Glacier Flexible Retrieval** after **90 days**.
5. Configured expiration of current versions after **365 days**.
6. Configured deletion of noncurrent versions after **91 days**.
7. Successfully created the lifecycle rule



6. Push some objects to S3 using the AWS CLI.

Installed and verified AWS CLI.

1. Configured the Devops profile.
2. Configured the AWS region.
3. Verified AWS identity using `aws sts get-caller-identity`.
4. Created `cli-test.txt`.
5. Uploaded the file using `aws s3 cp`.
6. Verified the object using `aws s3 ls`.

```
wuhaib@ZED_NOTE_CX3 MINGW64 ~/Downloads (master)
$ aws sts get-caller-identity --profile Devops
{
  "UserId": "AIDAYWQXDC5YM2UJQAOWG",
  "Account": "598122633072",
  "Arn": "arn:aws:iam::598122633072:user/Devops"
}

wuhaib@ZED_NOTE_CX3 MINGW64 ~/Downloads (master)
$ aws configure list --profile Devops
NAME      : VALUE                                : TYPE            : LOCATION
profile   : Devops                                : manual          : --profile
access_key : *****QXVC                          : shared-credentials-file :
secret_key : *****Laeb                          : shared-credentials-file :
region    : us-east-1                             : config-file      : ~/.aws/config

wuhaib@ZED_NOTE_CX3 MINGW64 ~/Downloads (master)
$ echo "Hello from AWS CLI" > cli-test.txt

wuhaib@ZED_NOTE_CX3 MINGW64 ~/Downloads (master)
$ cat cli-test.txt
Hello from AWS CLI

wuhaib@ZED_NOTE_CX3 MINGW64 ~/Downloads (master)
$ ws s3 cp cli-test.txt s3://YOUR-BUCKET-NAME/ --profile Devops
bash: ws: command not found

wuhaib@ZED_NOTE_CX3 MINGW64 ~/Downloads (master)
$ aws s3 cp cli-test.txt s3://syed-s3-lab-2026-001/ --profile Devops
upload failed: .\cli-test.txt to s3://syed-s3-lab-2026-001/cli-test.txt An error occurred (AccessDenied) when calling the PutObject action: User: arn:aws:s3:::syed-s3-lab-2026-001/cli-test.txt is not authorized to perform: s3:PutObject on resource: "arn:aws:s3:::syed-s3-lab-2026-001/cli-test.txt"

wuhaib@ZED_NOTE_CX3 MINGW64 ~/Downloads (master)
$ aws s3 cp cli-test.txt s3://syed-s3-lab-2026-001/ --profile Devops
upload: .\cli-test.txt to s3://syed-s3-lab-2026-001/cli-test.txt

wuhaib@ZED_NOTE_CX3 MINGW64 ~/Downloads (master)
$ aws s3 ls s3://syed-s3-lab-2026-001/ --profile Devops
2026-09-08 18:47:03      19 cli-test.txt
2026-09-08 16:59:40      14 file1.txt
2026-09-08 16:59:41      15 file2.txt
2026-09-08 17:37:10    734425 git part 2.docx
2026-09-08 17:06:44     190 index.html

wuhaib@ZED_NOTE_CX3 MINGW64 ~/Downloads (master)
$ |
```

7. Write a Bash script to create an S3 bucket.

1. Created a Bash script using the `vi` editor.
2. Defined bucket name, region, and AWS CLI profile.
3. Used `aws s3api create-bucket` to create the bucket.
4. Added success and failure conditions.
5. Made the script executable using `chmod`.
6. Executed the script successfully.

Verified the created bucket.

 MINGW64:/c/Users/wuhaib/Downloads

```
wuhaib@ZED_NOTE_CX3 MINGW64 ~/Downloads (master)
$ vi create-s3-bucket.sh

wuhaib@ZED_NOTE_CX3 MINGW64 ~/Downloads (master)
$ chmod +x create-s3-bucket.sh

wuhaib@ZED_NOTE_CX3 MINGW64 ~/Downloads (master)
$ ./create-s3-bucket.sh
Creating S3 bucket: syed-s3-bash-2026-001
{
  "Location": "/syed-s3-bash-2026-001",
  "BucketArn": "arn:aws:s3:::syed-s3-bash-2026-001"
}
Bucket created successfully!
Bucket details:
{
  "LocationConstraint": null
}

wuhaib@ZED_NOTE_CX3 MINGW64 ~/Downloads (master)
$
```

8 *Upload 1 GB File Using AWS CLI

- Created a 1 GB file named `largefile.bin`.
- Used the `dd` command to generate the file.
- Verified the file size.
- Uploaded the file using `aws s3 cp`.
- Verified the upload using `aws s3 ls`.
- Demonstrated handling of large-file uploads through AWS CLI.

```
wuhaib@ZED_NOTE_CX3 MINGW64 ~/Downloads (master)
$ ls -lh largefile.bin
-rw-r--r-- 1 wuhaib 197121 1.0G Sep  8 18:52 largefile.bin

wuhaib@ZED_NOTE_CX3 MINGW64 ~/Downloads (master)
$ aws s3 cp largefile.bin s3://syed-s3-lab-2026-001/ --profile Devops
Completed 1.0 GiB/1.0 GiB (2.7 MiB/s) with 1 file(s) remaining

cancelled: ctrl-c received

wuhaib@ZED_NOTE_CX3 MINGW64 ~/Downloads (master)
$ aws s3 ls s3://syed-s3-lab-2026-001/ --profile Devops
2026-09-08 18:47:03          19 cli-test.txt
2026-09-08 16:59:40          14 file1.txt
2026-09-08 16:59:41          15 file2.txt
2026-09-08 17:37:10       734425 git part 2.docx
2026-09-08 17:06:44          190 index.html

wuhaib@ZED_NOTE_CX3 MINGW64 ~/Downloads (master)
$ aws s3 cp largefile.bin s3://syed-s3-lab-2026-001/ --profile Devops
upload: .\largefile.bin to s3://syed-s3-lab-2026-001/largefile.bin

wuhaib@ZED_NOTE_CX3 MINGW64 ~/Downloads (master)
$ aws s3 ls s3://syed-s3-lab-2026-001/ --profile Devops
2026-09-08 18:47:03          19 cli-test.txt
2026-09-08 16:59:40          14 file1.txt
2026-09-08 16:59:41          15 file2.txt
2026-09-08 17:37:10       734425 git part 2.docx
2026-09-08 17:06:44          190 index.html
2026-09-08 19:00:37    1073741824 largefile.bin

wuhaib@ZED_NOTE_CX3 MINGW64 ~/Downloads (master)
$ |
```